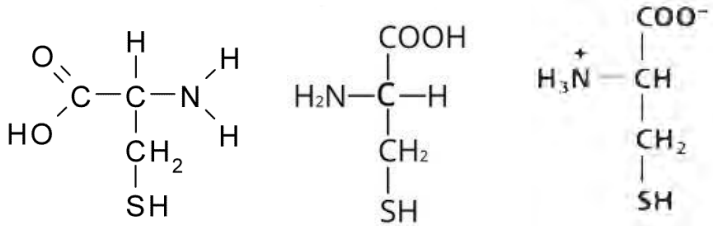


Question number	Answer	Additional guidance	Mark
5(a)(i)	A description that includes two of the following points: • long (chains of amino acids / polypeptides / proteins) (1) • repeating sequences of amino acids (1) • high proportion of {small / non-polar / hydrophobic} {R groups / amino acids} (1) • (parallel) chains held with {cross links / hydrogen bonds}	IGNORE large ACCEPT a description IGNORE named amino acids ACCEPT hydrophobic groups on the outside IGNORE stated number of chains / refs to triple helix DO NOT ACCEPT other named bonds	(2)

Question number	Answer	Additional guidance	Mark
5(a)(ii)	A diagram that includes the following points: • a central carbon atom (1) • an amino group and a carboxyl group (1) • a hydrogen and the cysteine R group (attached by the C) (1)		(3)

Question number	Answer	Additional guidance	Mark
5(b)(i)	A description that includes the following points: • β {pleated / sheet} / α helix (1) • (held together by) hydrogen bonds (between amino acids) (1)	DO NOT ACCEPT {double helix / triple helix} even if β sheet named IGNORE B / A DO NOT ACCEPT other bonds DO NOT ACCEPT between the R groups	(2)

Question number	Answer	Additional guidance	Mark
5(b)(ii)	An explanation that includes four of the following points: • primary structure remains the same length and secondary structure is longer with higher temperature (1) • (primary structure remains the same length) because the peptide bonds are not affected (by the heat) (1) • secondary structure is shorter than primary structure because of the {folding / winding} (1) • (secondary structure gets longer) because (the heat) hydrogen bonds break (1) • due to increase in vibration of the {R groups / amino acids / molecule} (1)	Piece together ACCEPT strong / hard to break ACCEPT increase in kinetic energy results in more movement within the molecule.	(4)